INFO-H3000 : Recherche opérationnelle

Introduction to Voting Theory

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Introduction

- Four main voting procedures
  - Plurality, Two-round, Borda, Condorcet
- Analysis and paradoxes
  - Properties
- Theoretical results: May's theorem & Arrow's impossibility theorem

Examples ???

Let A={a,b,c,d,. . . } be a set of candidates

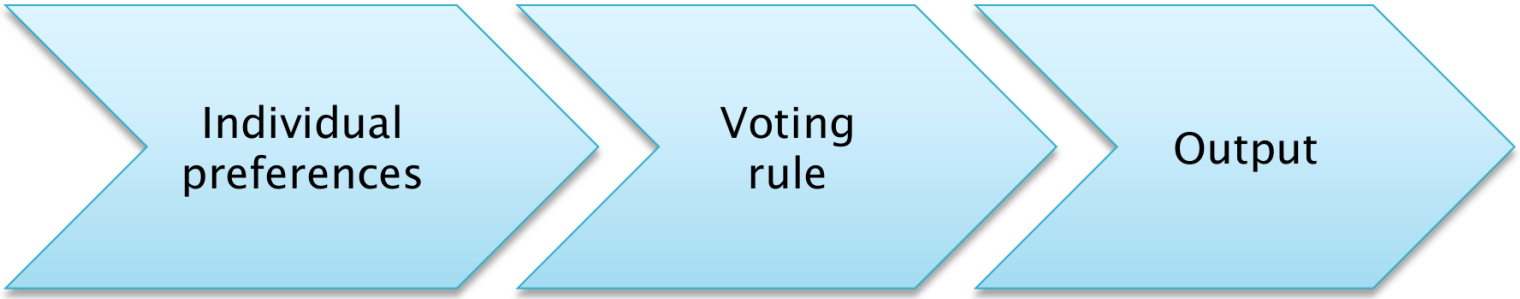
- There are N voters
- Each voter can express his preferences on the basis of a total order
  - Example: Si A={a,b,c,d}, b > a > c > d A total order on A is a binary relation that is transitive, antisymmetric and total

To merge the individual preferences into acceptable global (social) ones.

- What does it means ? What is acceptable ?
  - A candidate that is preferred to any other ones by a majority of voters is not elected ?[1]
  - A candidate that regresses in the voters' preferences becomes elected ?
  - The global preference between two given candidates might be affected by the presence of a third candidate ? *

[1] During the US elections in 2000, Gore received 50999897 votes, Bush 50456002

Four main voting procedures



Order ?

Favorite candidate ?

Stable ?

Public ?

Solution existence ?

Order ?

Ties ?

Favorite candidate ?

Plurality voting

- Every voter communicates his preferred candidate;
- The candidate that received the more number of votes is elected.
- Example:
  - For 11 voters: $a > b > c$
  - For 8 voters: $b > a > c$
  - For 2 voters: $c > b > a$

Two-round voting

- Every voter communicates his preferred candidate;
- The two candidates that received the more number of votes are selected;
- Every voter communicates his preferred candidate among the two selected candidates;
- The candidate that received the more number of votes is elected.
- Example:
  - For 11 voters: $a > b > c$
  - For 8 voters: $b > a > c$
  - For 2 voters: $c > b > a$

Borda's rule

- Let us assume that there are k candidates;
- Every voter communicates his ranking;
- Rank 1 candidates received k points;
- Rank 2 candidates received k-1 points;
- ...
- Rank k candidates receive 1 points;
- Borda score = sum of points
  - For 11 voters: a > b > c
  - For 8 voters: b > a > c
  - For 2 voters: c > b > a

Condorcet's rule

- Idea: pairwise comparisons
- Let n_{ij} be the number of voters who prefer candidate i to candidate j. Obviously, we have que $n_{ij} + n_{ji} = N$
- A candidate i is globally preferred to another candidate j iff $n_{ij} > N/2$ (if N is odd)
- Example:
  - For 11 voters: a > b > c
  - For 8 voters: b > a > c
  - For 2 voters: c > b > a

How to choose between these procedures

An embarrassing example

Exemple: N = 27

- For 5 voters x > y > z > t
- For 4 voters x > z > y > t
- For 2 voters t > y > x > z
- For 6 voters t > y > z > x
- For 8 voters z > y > x > t
- For 2 voters t > z > y > x

With plurality voting

- For 5 voters $x > y > z > t$
- For 4 voters $x > z > y > t$
- For 2 voters $t > y > x > z$
- For 6 voters $t > y > z > x$
- For 8 voters $z > y > x > t$
- For 2 voters $t > z > y > x$
- « Scores » $t = 10$, $x = 9$, $z = 8$, $y = 0$
- We conclude $t > x > z > y$
- Nevertheless: 17 voters prefer x to t !!!

Two-round voting

- For 5 voters $x > y > z > t$
- For 4 voters $x > z > y > t$
- For 2 voters $t > y > x > z$
- For 6 voters $t > y > z > x$
- For 8 voters $z > y > x > t$
- For 2 voters $t > z > y > x$
- « Scores » $t = 10$, $x = 9$, $z = 8$, $y = 0$
- First round: x and t are selected
- Second round: x wins !
- Nevertheless: 18 voters prefer y to x !!!

Borda's rule

- For 5 voters x > y > z > t
- For 4 voters x > z > y > t
- For 2 voters t > y > x > z
- For 6 voters t > y > z > x
- For 8 voters z > y > x > t
- For 2 voters t > z > y > x
- For x we have a 5*4+ 4*4 +2*2 +6*1+8*2+2*1 =64
- Score for y = 75, score for z = 74, score for t = 57
- Y wins and we may conclude y > z > x > t
- Nevertheless:14 voters prefer z to y

Condorcet's rule

- For 5 voters x > y > z > t
- For 4 voters x > z > y > t
- For 2 voters t > y > x > z
- For 6 voters t > y > z > x
- For 8 voters z > y >x > t
- For 2 voters t > z > y > x
- $n_{xt} = 17, n_{yx} = 18, n_{yt} = 17, n_{zx} = 16, n_{zy} = 14, n_{zt} = 17$

To conclude:

- Plurality voting => t
- Two-round voting => x
- Borda's rule => y
- Condorcet method => z
- Choosing a method = choosing a solution !

How to characterize voting procedures ?

Unanimity, monotonicity, dependance to third alternatives(manipulation) , condorcet criterion, solution existence, separability ?

Monotonicity

- Example: (N=17, A={x,y,z}):
  - For 6 voters: x > y > z
  - For 5 voters: z > x > y
  - For 4 voters: y > z > x
  - For 2 voters: y > x > z
- First round: x and y are selected
- Second round: x wins

Two-round voting

- x convince the two last voters: they swap their preferences between x and y
- Example:
  - For 6 voters: $x > y > z$
  - For 5 voters: $z > x > y$
  - For 4 voters: $y > z > x$
  - For 2 voters: $x > y > z$

  Z wins!!!

Dependence to third alternatives/manipulation

Borda's rule

- Example: N=7, A={a,b,c,d}
  - For 3 voters: c > b > a > d
  - For 2 voters: b > a > d > c
  - For 2 voters: a > d > c > b
- We have a > b > c > d
- Candidate d withdrawes (anyway he has no chance to be elected)
- We have c b a !!! The withdrawal of the worst candidate might have a strong effect on the ranking !

Adding a candidate

- Example: N=5, A={a,b,c}
  - For 2 voters: a > b > c
  - For 2 voters: b > c > a
  - For 1 voter: c > a > b
- Scores: a(9), b(11), c(9)

Adding a candidate

- Example: N=5, A={a,b,c,d}
  - For 2 voters: a > d > b > c
  - For 2 voters: b > c > a > d
  - For 1 voter: c > a > b > d
- Scores: a(15), b(14), c(10), d(9)

What about Condorcet's rule ?

- For 11 voters: a > b > c
- For 8 voters: b > a > c
- For 2 voters: c > b > a
  - BORDA: a b c
- For 11 voters: a > b > c
- For 8 voters: b > c > a
- For 2 voters: c > b > a
  - BORDA: b a c
- For 11 voters: a > c > b
- For 8 voters: b > c > a
- For 2 voters: c > b > a
  - BORDA: c a b

But

- For 11 voters: a > b > c
- For 8 voters: b > a > c
- For 2 voters: c > b > a

The supporters of b decide to put a disadvantage

- For 11 voters: a > b > c
- For 8 voters: b > c > a
- For 2 voters: c > a > b

Two-round voting

- Example : A={a,b,c,d}, N=21
  - For 10 voters: b > a > c > d
  - For 6 voters: c > a > d > b
  - For 5 voters: a > d > b > c
- First round: b and c
- Second round: b wins

Two-round voting

- The six voters who « hate » b decide to modify their vote
  - For 10 voters: $b > a > c > d$
  - For 6 voters: $c > a > d > b$
  - For 5 voters: $a > d > b > c$
- First round: b and a
- Second round: a wins

Abstention TRV

- Example : A=a,b,c, N=11
  - For 4 voters: a > b > c
  - For 4 voters: c > a > b
  - For 3 voters: b > c > a
- First round: a and c
- Second round: c wins

Abstention TRV

- 2 of the 4 first voters exptect c to be elected; they decide to withdraw...
  - For 2 voters: a > b > c
  - For 4 voters: c > b > a
  - For 3 voters: b > c > a
- First round: c and b
- Second round: b wins

Solution existence ?

Condorcet's rule

A={x,y,z}, N=60 voters

- For 23 voters: $x > y > z$
- For 17 voters: $y > z > x$
- For 2 voters: $y > x > z$
- For 10 voters: $z > x > y$
- For 8 voters: $z > y > x$

Separability

Two-round voting

Example: 3 candidates, 2 regions, 13 voters per region

Region 1

For 4 voters: $a > b > c$
For 3 voters: $b > a > c$
For 3 voters: $c > a > b$
For 3 voters: $c > b > a$

Round 1	
a	4
b	3
c	6

Round 2	
a	7
c	6

Region 2

For 4 voters: $a > b > c$
For 3 voters: $b > a > c$
For 3 voters: $c > a > b$
For 3 voters: $b > c > a$

Round 1	
a	4
b	6
c	3

Round 2	
a	7
b	6

If both regions were simultaneously considered

- First round ! A:8, B:9 and C:9
- Second round ! B:17 et C:9
- Non separability

Overview

	Plurality	Two round	Borda	Condorcet
Monotonicity	?	✗	?	?
Solution existence	✓	✓	✓	✗
Independence to third candidates	?	✗	✗	✗
Separability	?	✗	?	?
Condorcet criterion	✗	✗	✗	✓

Other methods ???

Sequential voting

(aka pairwise sequential plurality voting)

- For example A = a,b,c,d, N= 3 voters
  - For 1 voter: a > b > c > d
  - For 1 voter: b > c > d > a
  - For 1 voter: c > d > a > b
- A first given sequence: (((a,b),c),d) ! d wins
- ... but a is preferred by all the voters (unanimity !)
- A second possible sequence (((d,c),b),a) ! b wins !!!

Modified Borda's rule

- Points are only assigned to the k first candidates:
  - Pour 3 voters: x > y > z > t
  - Pour 1 voters: y > z > x > t
  - Pour 1 voters: y > z > t > x
  - Pour 2 voters: z > t > x > y
- If k=1, we have x(3), y(2), z(2), t(0)
- If k=2, we have y(7), x(6), z(6), t(2)
- If k=3, we have z(13), x(12), y(12), t(5) 45

Some theoretical results

May's theorem(1952)

- Desirable properties ?
  - Neutrality
  - Anonimity
  - Monotonicity
- Main results: With 2 candidates, the only neutral, anonymous and monotonic voting procedure is the plurality rule !

Arrow's impossibility theorem

Arrow's theorem

« A Difficulty in the Concept of Social Welfare », The Journal of Political Economy, vol. 58 (4), 1950

We consider:

- Three candidates: x, y and z
- x'
- Two voters
- P_i preference relation
- I_i indifference relation
- For any couple x and y we have x P_i y, x I_i y or x P_i y
- $S_i = P_i \cup I_i$

Arrow's axioms:

- Axiom 1: $\forall$ x',y', we have: x' S y' or y' S x'
- Axiom 2: $\forall$ x',y',z', we have: x' S y', y' S z' $\rightarrow$ x' S z'
- Obviously we have:
  - x' P y' $\equiv$ x' S y' et y' $\sim$ S x'
  - x' I y' $\equiv$ x' S y' et y' S x'

Lemma:

- ∀ x', x' S x'
- If x' P y' ! x' S y'
- If x' P y', y' P z' ! x' P z'
- If x' I y', y' I z' ! x' I z'
- ∀ x',y' we have: x' S y' or y' P x'
- If x' P y', y' S z' ! x' P z'

Arrow's conditions

- Condition 1: The social welfare function is defined for every admissible pair of individual orderings R_1 and R_2
- Condition 2: If an alternative social state rises or does not fall in the ordering of each individual without any other change in those orderings and if x was preferred to another alternative y before the change of individual orderings, then x is still preferred to y
- Condition 3: Let R_1 and R_2 and R_1' and R_2' be two sets of individual orderings. If, for both individuals i and for all x and y in a given set of alternatives S, x R_i y if and only if x R_i' y, then the social choice made from S is the same wether the individual orderings are R_1, R_2 or R_1', R_2'

Arrow's conditions

- Definition: A social welfare function will be said to be imposed if for some pair of distinct alternatives x and y, x R y for any set of individual orderings R_1 et R_2, where R is the social ordering corresponding to R_1 et R_2
- Condition 4: The social welfare function is not imposed.

Arrow's conditions

- Definition: A social welfare function is said tobe « dictatorial » if there exists an individual isuch that for all x' and y', x' Pi y' implies x' Py' regardless the orderings of all individualsother than i, where P is the social preferencerelation corresponding to those orderings.
- Condition 5: The social welfare function isnot to be dictatorial.

Consequences of Arrow's conditions:

- Consequence 1: If $x' P_1 y'$ and $x' P_2 y'$, then $x' P y'$
- Consequence 2: Suppose that for some x' and y', whenever $x' P_1 y'$ and $y' P_2 x'$, $x' P y'$. Then, for that x' and y', whenever $x' P_1 y', x' P y'$.
- Consequence 3: If $x' P_1 y'$ and $y' P_2 x'$, then $x' I y'$

The consequences of the consequences:

- Let us assume R1: x,y,z et R2: z,x,y

Anonimity / Equality ?

- Are there systems for which voters may be discriminated ?
- Number of shares in companies
- Example: the selection of a vice-president
- Weights: $[103 : 101, 97, 2]$

Shareholder	**Shares**
Doug	101
Nicolas	97
Elisabeth	2

Is Nicolas 48 times more important than Elisabeth?
Can we quantify corruption?

The Banzhaf Power Index (1965)

- John F. Banzhal, lawyer & professor
- Series of lawsuits against Tobacco and fastfood industries
- A voter in a winning coalition is said to be critical if the voter's withdrawal from the coalition would cause it to be become a losing coalition

The Banzhaf Power Index (1965)

- The Banzhaf power of a voter is the number of winning coalitions in which the voter is critical
- The total Banzhaf power of a system is the sum of the Banzhaf powers of all voters in the system
- The Banzhaf index of a voter is the Banzhaf power of the voter divided by the total Banzhal power of the system

The board of supervisors of Nassau County, New York(1965)

Hempstead 1	31	
Hempstead 2	31	
Oyster Bay	28	Q=58 (115)
North Hempstead	21	
Long Beach	2	
Glen Cove	2	

- 32 winning coalitions
- 3 have 2 members, 10 have 3 members, 12 have 4 members, 6 have five members, 1 has six members
- A total of 48 critical voters
- Conclusion: The power of the board is equally distributed among the three largest district
- Combinatorial explosion...

Parliamentary representations

- We consider a population of n individuals
- Let $P_1, P_2, \ldots P_N$ be a partition this population into N classes. The size of P_i is denoted p_i
- GOAL: to elect S representatives
- Quota of P_i is given by: $q_i = S.\frac{p_i}{n}$

Hamilton's rule

$$(s_1, s_2, \ldots, s_N) \in \mathbb{N}^N \quad \Big| \quad \sum_{i=1}^{N} s_i = S$$

$$\forall i \in \{1, 2, \ldots, N\} : \quad s_i = \lfloor q_i \rfloor \text{ ou } \lfloor q_i \rfloor + 1,$$

$$\forall i : \; s_i = \lfloor q_i \rfloor + 1, \; \forall j : \; s_j = \lfloor q_j \rfloor : \quad q_i - \lfloor q_i \rfloor \geq q_j - \lfloor q_j \rfloor.$$

Example

	p_i	q_i	$\lfloor q_i \rfloor$	s_i
P1	6373	6,373	6	6
P2	2505	2,505	2	2
P3	602	0,602	0	1
P4	520	0,520	0	1
total	10000		8	10

	p_i	q_i	$\lfloor q_i \rfloor$	s_i
P1	6373	6,373	6	6
P2	2512	2,512	2	**?**
P3	603	0,603	0	1
P4	512	0,512	0	**?**
total	10000		8	10

Monotonicity

$S = 7$

	p_i	q_i	$[q_i]$	s_i
P1	20	1,373	1	2
P2	34	2,333	2	2
P3	48	3,294	3	3
total	102		6	7

$S = 8$

	p_i	q_i	$[q_i]$	s_i
P1	20	1,569	1	1
P2	34	2,667	2	3
P3	48	3,765	3	4
total	102		6	8

Alabama Paradox

Even worse

- One more votes for P1

	p_i	q_i	$[q_i]$	s_i
P1	21	1,631	1	1
P2	34	2,641	2	3
P3	48	3,728	3	4
total	103		6	8

Population paradox

Initial population

	p_i	q_i	$[q_i]$	s_i
P1	1786	9,654	9	10
P2	1243	6,719	6	7
P3	671	3,627	3	3
total	3700		18	20

After adding 65 voters

	p_i	q_i	$[q_i]$	s_i
P1	1805	9,588	9	9
P2	1283	6,815	6	7
P3	677	3,596	3	4
total	3765		18	20

65 voters have been forgotten!
19 for P1, 40 for P2 and **only** 6 for P3

Population transfert paradox

	p_i	q_i	$[q_i]$	s_i
P1	14	2,593	2	3
P2	10	1,852	1	2
P3	3	0,556	0	0
total	27		3	5

	p_i	q_i	$[q_i]$	s_i
P1	14	2,593	2	2
P2	9	1,667	1	2
P3	4	0,741	0	1
total	27		3	5

Jefferson's method

$$(s_1, s_2, \ldots, s_N) \in \mathbb{N}^N \quad \Big| \quad \sum_{i=1}^{N} s_i = S$$

$$\forall i \in \{1, 2, \ldots, N\} : \; s_i \neq 0, \; \frac{p_i}{s_i} \geq \frac{p_j}{s_j + 1}, \; \forall j \in \{1, 2, \ldots, N\}$$

Jefferson's method

Proposition: $s_i \geq [q_i]$

Obvious if $[q_i] = 0$.
If $[q_i] \neq 0$. We have:

$$\frac{p_i}{[q_i]+1} < \frac{p_i}{q_i} < \frac{p_i}{[q_i]} \quad \text{and} \quad \frac{p_i}{q_i} = \frac{n}{S}$$

If $s_i < [q_i]$ then $s_j \leq [q_j]$. Imagine $\exists j | s_j > [q_j]$:

$$\frac{p_j}{s_j} < \frac{n}{S} \leq \frac{p_i}{s_i+1}.$$

As a consequence :

$$\sum_{i=1}^{N} s_i < \sum_{i=1}^{N} [q_i] < S \text{ !!!}$$

Example 1

	pi	qi	[qi]	pi/([qi]+1)	pi/([qi]+2)	si
P1	6373	6,4	6	910,42857	796,625	7
P2	2505	2,5	2	835	626,25	3
P3	602	0,6	0	602	301	0
P4	520	0,5	0	520	260	0
total	10000		8			10

Example 2

	pi	qi	[qi]	pi/([qi]+1)	pi/([qi]+2)	si
P1	6373	6,4	6	910,42857	796,625	8
P2	2305	2,3	2	768,33333	576,25	2
P3	702	0,7	0	702	351	0
P4	620	0,6	0	620	310	0
total	10000		8			10

Consistency

$$p_i < p_j \Rightarrow s_i \leq s_j$$

Population transfert paradox

	pi	qi	[qi]	pi/([qi]+1)	pi/([qi]+2)	si
P1	936	2,3	2	312	234	3
P2	623	1,5	1	311,5	207,6667	1
P3	488	1,2	1	244	162,6667	1
total	2047		4			

	pi	qi	[qi]	pi/([qi]+1)	pi/([qi]+2)	si
P1	936	2,3	2	312	234	2
P2	626	1,5	1	313	208,6667	2
P3	485	1,2	1	242,5	161,6667	1
total	2047		4			

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